From Attribute Alignment to Attribute Verification: GARNET for Text-Based Person Retrieval

Sachin Kumar Bairwa
d24145@students.iitmandi.ac.in

Vandita Dutt
d22140@students.iitmandi.ac.in

Pranay
s24015@students.iitmandi.ac.in

Dinesh Singh
dineshsingh@iitmandi.ac.in

Indian Institute of Technology Mandi, India

Abstract

Text-based person retrieval (TBPR) has progressed from global image–text matching to fine-grained phrase–region alignment, attribute-aware modeling, and perturbation-sensitive learning. These capabilities, however, do not by themselves guarantee that a specific changed attribute *is verified against its corresponding local visual evidence. An otherwise correct description can therefore remain compatible with an image even when one decisive attribute is changed, because the corrupted phrase is not explicitly required to lose visual support.*

We introduce ***GARNET****, a grounded counterfactual attribute-verification framework that explicitly couples semantic corruption with local visual evidence. GARNET combines* ***Phrase–Patch Grounding (PPG)****, which exposes phrase-specific support over image patches, with* ***Counterfactual Attribute Verification (CAV)****. CAV-G requires the correct caption to outrank a controlled single-attribute corruption, while CAV-L requires the original changed phrase to receive stronger local support than its corrupted counterpart. Thus, GARNET moves from attribute correspondence or generic perturbation sensitivity toward explicit* grounded attribute verification.

We evaluate standard retrieval on CUHK-PEDES, ICFG-PEDES, and RSTPReid, and use controlled verification diagnostics on CUHK-PEDES to directly test the targeted failure mode. GARNET remains competitive under conventional retrieval while reaching 91.92% *Local C>C, improving the strongest evaluated prior local baseline by* 10.40 *points, with a signed local gap of* $\Delta_\ell = 0.135$.

1. Introduction

Text-based person retrieval (TBPR) aims to retrieve the pedestrian image that best matches a free-form natural-language description. The task is inherently fine-grained: different identities often share most of their appearance, while the decisive distinction may be a small semantic detail such as *blue shirt* versus *red shirt*, *backpack* versus *handbag*, or *short sleeves* versus *long sleeves*. A successful model must therefore determine not only whether an image and a caption are globally compatible, but also whether the specific attributes expressed in the query are supported by visual evidence.

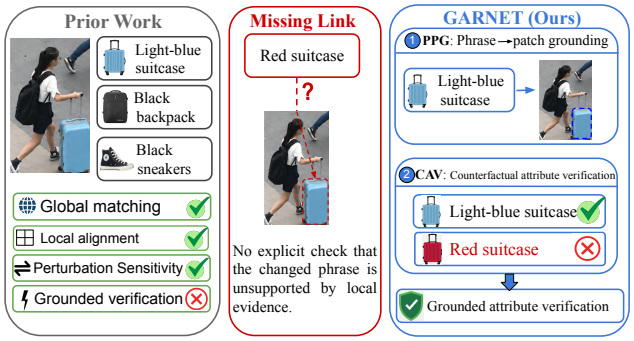


Figure 1. **Motivation for GARNET.** Prior TBPR methods have advanced global matching, fine-grained attribute correspondence, and perturbation sensitivity. GARNET targets the remaining missing connection: a controlled changed attribute is explicitly tied to its local visual evidence and is required to lose support when it becomes visually inconsistent.

TBPR has progressed substantially from global cross-modal matching toward richer visual–language representations. CMPM/CMPC [18] learns discriminative shared image–text embeddings, CLIP [10] provides transferable vision–language representations, and IRRA [5] combines similarity learning with implicit cross-modal relation reasoning. These approaches provide strong global compatibility, but a high image–caption score does not explicitly guarantee that every fine-grained attribute is visually correct.

A large body of work therefore models local and attribute-aware correspondence. ViTAA [14] explicitly aligns visual

and textual attributes, UFineBench/CFAM [21] studies ultra-fine-grained matching, MARS/MARS++ [4] emphasizes adjective–noun visual attributes, and KPDM [17] models key phrases together with image-patch importance. QC-Align [6] and CECA [20] further strengthen structured fine-grained correspondence. Thus, the issue addressed here is *not* that prior TBPR ignores attributes or local alignment.

A complementary direction improves sensitivity to modified or difficult text. RaSa [1] explicitly learns sensitivity to replaced words, while hard-negative learning is also used for fine-grained discrimination [21]. Hence, the limitation is also not simply that prior models ignore textual perturbations. Detecting that a caption has changed does not necessarily establish *which local visual evidence* should invalidate the changed semantic content.

Figure 1 illustrates the remaining gap. Consider an otherwise correct description in which only *light-blue suitcase* is changed to *red suitcase*. Most of the caption can remain compatible with the same image. A global matcher may retain a high caption-level score, a fine-grained model may still align the unchanged attributes correctly, and a perturbation-sensitive objective may recognize that the text has been modified. The critical question is whether the model explicitly verifies that the changed phrase *red suitcase* is unsupported by the visual evidence of the suitcase itself. This motivates the specific problem addressed by GARNET. To our knowledge, existing TBPR objectives do not explicitly couple a controlled single-attribute semantic corruption to the local visual evidence of the exact changed phrase and require that evidence to decrease when the phrase becomes visually inconsistent. Prior work has substantially advanced *attribute correspondence* and *perturbation sensitivity*; the missing requirement is an explicit connection between *what semantic attribute changed* and *whether the corresponding grounded evidence still supports it*.

We propose **GARNET**, a grounded counterfactual attribute-verification framework. **Phrase–Patch Grounding (PPG)** exposes phrase-specific visual evidence by matching attribute phrases to local image patches. **Counterfactual Attribute Verification (CAV)** then verifies controlled single-attribute counterfactuals at two levels: CAV-G requires the correct caption to score above its corrupted counterpart, while CAV-L requires the original changed phrase to receive stronger local support than the corrupted phrase. GARNET therefore connects *where an attribute is supported* with *whether that support decreases when the attribute becomes semantically incorrect*. We additionally use MALS person-domain pre-training [15] as a representation-strengthening stage and isolate its effect from PPG and CAV in the ablation study.

Our contributions are summarized as follows:

- We formulate and empirically characterize a **grounded attribute-verification gap** in TBPR: fine-grained correspondence or perturbation sensitivity does not necessarily require a controlled wrong attribute to lose the local visual support of the exact changed phrase.
- We introduce **GARNET**, which couples **Phrase–Patch Grounding (PPG)** with **Counterfactual Attribute Verification (CAV)** so that controlled semantic contradictions are rejected both globally and at the changed phrase.
- We evaluate conventional retrieval on CUHK-PEDES, ICFG-PEDES, and RSTPReid, and introduce controlled CUHK-PEDES diagnostics including attribute-confusable retrieval, global/local correct-over-corrupt accuracy, signed corruption gaps, and paraphrase robustness to directly test the behavior targeted by GARNET.

2. Related Work

2.1. Global and Implicit Cross-Modal Alignment

Text-based person retrieval requires bridging the modality gap between free-form descriptions and pedestrian images. Early work therefore emphasized discriminative shared representations and global image–text matching. CMPM/CMPC [18] learns cross-modal projection matching and classification, while CLIP [10] provides transferable vision–language representations. IRRA [5] further combines Similarity Distribution Matching with masked language modeling and implicit cross-modal relation reasoning.

These methods substantially improve overall image–caption compatibility, but global alignment does not explicitly determine whether every individual attribute in the description is visually supported. When two captions differ in only one decisive attribute, most semantic content can remain compatible with the same image; therefore, strong global matching alone does not ensure that the changed attribute is the reason a corrupted description is rejected.

2.2. Fine-Grained, Phrase-Level, and Attribute-Aware Alignment

A large body of work models local, part-level, phrase-level, or attribute-level correspondence. ViTAA [14] learns attribute-specific visual–textual alignment, SSAN [3] extracts semantically aligned part-level features, and LGUR [12] addresses visual–textual granularity discrepancy. UFineBench/CFAM [21] studies ultra-fine-grained matching, MARS/MARS++ [4] emphasizes adjective–noun visual attributes, and KPDM [17] exploits key phrases together with image-patch importance. QC-Align [6] formulates structured many-to-many word–region alignment, while CECA [20] improves detailed correspondence through conversational attribute mining.

This literature is important for positioning GARNET because it shows that fine-grained alignment and attribute modeling are already well studied. Our claim is not that prior methods fail to localize attributes. Rather, to our knowledge,

the methods considered above do not explicitly couple a *known controlled semantic corruption* to the local visual evidence of the exact changed phrase and require that evidence to decrease when the phrase becomes visually inconsistent.

Person-domain pre-training provides an additional source of representation strength. APTM/MALS [15] demonstrates large-scale person–text pre-training with rich attribute supervision. GARNET uses MALS for person-domain initialization, but treats this stage as representation strengthening rather than the verification mechanism itself; its contribution is isolated from PPG and CAV experimentally.

2.3. Perturbation Sensitivity, Hard Negatives, and Robust Correspondence

A complementary direction exposes models to difficult, modified, or unreliable image–text relationships. RaSa [1] is particularly relevant because its Sensitivity-Aware objective encourages detection of replaced words. Hard-negative learning is also used to improve fine-grained discrimination [21]. RDE [8] addresses a related but different setting by learning robust visual–semantic correspondence under noisy image–text pairs.

These approaches improve sensitivity or robustness, but detecting a replaced word, ranking a hard negative lower, or identifying an unreliable pair does not necessarily specify which grounded visual evidence should invalidate the exact semantic change. The evaluated literature therefore leaves a gap at the intersection of perturbation awareness and local attribute correspondence.

Taken together, prior work has substantially advanced *global image–text matching*, *fine-grained attribute correspondence*, and *perturbation sensitivity*. GARNET targets their missing coupling: given a controlled counterfactual that changes exactly one visually verifiable attribute, the model should reduce not only the caption–image compatibility but also the local visual support of the *exact changed phrase*. PPG exposes phrase-specific evidence and CAV requires that controlled contradiction to be rejected at global and local levels, moving from correspondence or generic sensitivity toward explicit *grounded attribute verification*.

3. Methodology

3.1. Overview and Design Rationale

Scalars are written in lowercase italic, vectors in bold lowercase, matrices in bold uppercase, and sets in calligraphic type; images and text descriptions are denoted by uppercase italic symbols such as I and T.

Text-based person retrieval (TBPR) requires matching a natural-language description to the correct person image. Existing methods have made strong progress in global image–text alignment and relation learning [5], fine-grained or attribute-aware correspondence [3, 12–14], and perturbation-aware or hard-negative learning [1, 8, 21]. We therefore do *not* assume that prior TBPR ignores local attributes or textual perturbations. Rather, the remaining limitation lies at their intersection: local alignment establishes where textual details correspond, while perturbation-aware objectives encourage sensitivity to modified descriptions, but a controlled semantic attribute change is not explicitly tied to the local visual evidence of that same changed attribute.

Consider an otherwise correct description in which only "blue shirt" is changed to "red shirt". Most of the caption remains compatible with the image, so the corrupted description may still retain a high global matching score unless the changed attribute is explicitly verified against its visual evidence. GARNET is designed around this grounded verification requirement. Let

$$\mathcal{B} = \{(I_i, T_i, y_i)\}_{i=1}^{B} \tag{1}$$

denote a mini-batch of B image–caption pairs, where I_i is an image, T_i its description, and y_i the person identity. For the k-th single-attribute counterfactual $\widetilde{T}_{ik}$, let c_{ik} denote the index of the phrase modified in caption i. The desired behavior is

$$s_{\mathrm{g}}(I_i, T_i) > s_{\mathrm{g}}(I_i, \widetilde{T}_{ik}), \qquad a(I_i, p_{i,c_{ik}}) > a(I_i, \widetilde{p}_{i,c_{ik}}), \tag{2}$$

where $s_{\mathrm{g}}(\cdot,\cdot) \in \mathbb{R}$ is a global image–text similarity and $a(\cdot,\cdot) \in \mathbb{R}$ is the visual support of one phrase in one image. The first constraint requires the correct caption to outrank its counterfactual globally; the second requires the specific changed phrase to lose local support. GARNET realizes Eq. (2) through two problem-specific modules. **Phrase–Patch Grounding (PPG)** exposes phrase-specific visual evidence directly from the retrieval encoder, while **Counterfactual Attribute Verification (CAV)** verifies controlled attribute changes at global and local levels. Thus, PPG answers *where a phrase is visually supported*, and CAV asks *whether that support decreases when the phrase is made semantically incorrect*. CLIP initialization, SDM, MLM, identity supervision, person-domain pre-training, and EMA provide the general TBPR optimization backbone. Figure 2 summarizes the resulting architecture. The figure uses a concrete example for readability, whereas the formulation below is kept symbolic and applies to arbitrary numbers of phrases, patches, counterfactuals, and batch elements.

3.2. Phrase Extraction and Counterfactual Construction

For each caption T_i, we extract a set of visually meaningful phrases

$$\mathcal{P}_i = \{p_{ir}\}_{r=1}^{P_i}, \qquad 1 \le P_i \le P_{\max}, \tag{3}$$

where P_i is the number of valid phrases in caption i and $P_{\max}$ is an implementation-dependent upper bound. We

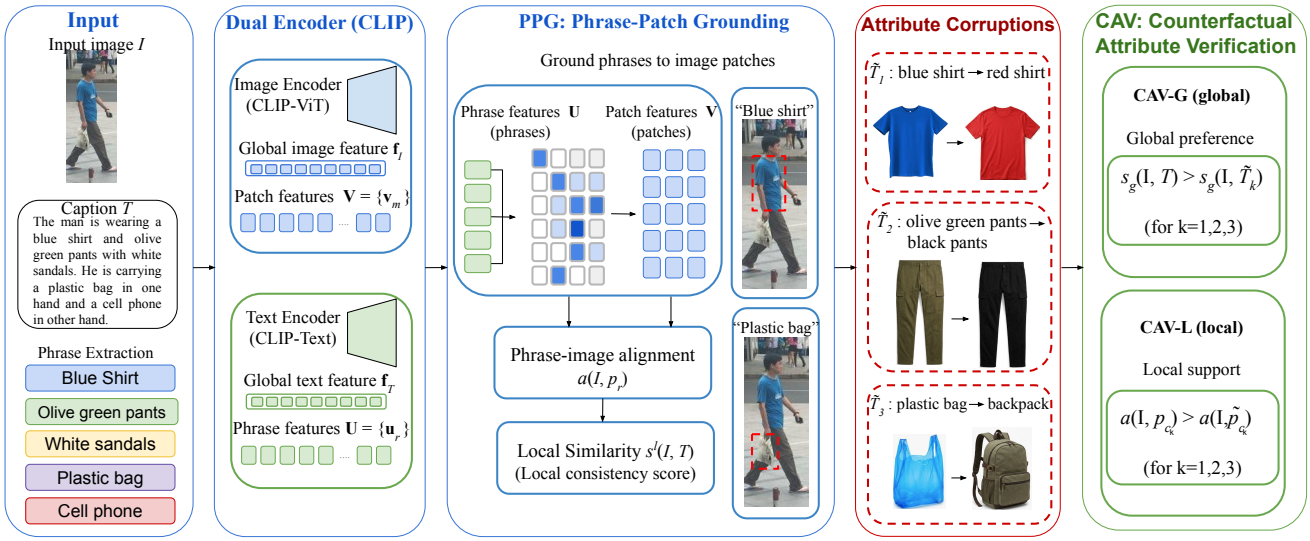


Figure 2. **GARNET architecture.** PPG grounds attribute phrases to visual patches and produces phrase-specific support together with an aggregated local similarity. Controlled single-attribute counterfactuals are then verified by CAV: CAV-G requires the correct caption to outrank its corrupted counterpart globally, while CAV-L requires the original changed phrase to retain stronger local visual support than its corrupted counterpart.

store the token span of each phrase under the tokenizer used by the text encoder. Meaning-preserving paraphrases are also cached for robustness augmentation using the same deterministic rule-based procedure employed at evaluation. The procedure uses controlled synonym substitution and light clause reordering; no external generative model is required. For each caption, we construct a set of K controlled counterfactuals,

$$\widetilde{\mathcal{T}}_i = \{\widetilde{T}_{ik}\}_{k=1}^{K}, \tag{4}$$

where every $\widetilde{T}_{ik}$ changes exactly one visually verifiable attribute while preserving the remainder of the description. The corruption may affect properties such as color, clothing type, carried object, or pose/action. We record

$$c_{ik} \in \{1, \ldots, P_i\}, \tag{5}$$

the index of the phrase changed by counterfactual k, together with the original phrase $p_{i,c_{ik}}$ and its corrupted counterpart $\widetilde{p}_{i,c_{ik}}$. Consequently, a counterfactual is not an arbitrary mismatched sentence: it differs from the original caption by one known semantic change, which allows CAV to test that change directly.

3.3. Dual-Encoder Representation

We use a CLIP-style dual encoder [10] and fine-tune both modalities end-to-end. Let the image encoder map image I_j to a normalized global representation

$$\mathbf{f}_j^I \in \mathbb{R}^d \tag{6}$$

and a matrix of normalized patch representations

$$\mathbf{V}_j = [\mathbf{v}_{j1}, \ldots, \mathbf{v}_{jN_p}]^\top \in \mathbb{R}^{N_p \times d}, \qquad \mathbf{v}_{jm} \in \mathbb{R}^d, \tag{7}$$

where N_p denotes the number of visual patches and d the shared image–text embedding dimension. The exact image resolution, patch grid, and backbone variant are implementation choices and are reported separately in the experimental setup. For caption T_i of token length L_i, the text encoder produces the contextual token-feature matrix

$$\mathbf{H}_i = [\mathbf{h}_{i1}, \ldots, \mathbf{h}_{iL_i}]^\top \in \mathbb{R}^{L_i \times d_t}, \qquad \mathbf{h}_{it} \in \mathbb{R}^{d_t}, \tag{8}$$

where d_t is the hidden dimension of the text transformer. The normalized global caption representation is

$$\mathbf{f}_i^T \in \mathbb{R}^d. \tag{9}$$

The scalar global similarity between caption T_i and image I_j is

$$s_{ij}^{\mathrm{g}} = s_{\mathrm{g}}(I_j, T_i) = (\mathbf{f}_i^T)^\top \mathbf{f}_j^I \in \mathbb{R}. \tag{10}$$

Collecting all pairwise similarities gives

$$\mathbf{S}^{\mathrm{g}} = [s_{ij}^{\mathrm{g}}]_{i,j=1}^{B} \in \mathbb{R}^{B \times B}. \tag{11}$$

Following identity-aware similarity distribution matching used in TBPR [5], we define the scalar target entry

$$q_{ij} = \frac{\mathbf{1}[y_i = y_j]}{\sum_{n=1}^{B} \mathbf{1}[y_i = y_n]}, \tag{12}$$

and collect the entries into $\mathbf{Q} = [q_{ij}]_{i,j=1}^{B} \in \mathbb{R}^{B \times B}$. Bidirectional SDM is then applied to $\mathbf{S}^{\mathrm{g}}$ and $\mathbf{Q}$, yielding $\mathcal{L}_{\mathrm{sdm}}$.

3.4. Phrase–Patch Grounding (PPG)

Global matching does not reveal which local visual evidence supports an individual phrase. PPG therefore derives phrase-level representations from the contextual token features and

aligns them directly to image patches. For phrase p_{ir} spanning token indices $[b_{ir}, e_{ir})$, let

$$\mathbf{W}_t \in \mathbb{R}^{d \times d_t} \tag{13}$$

denote the text projection into the shared embedding space. The normalized phrase vector is

$$\mathbf{u}_{ir} = \mathrm{norm}\left(\mathbf{W}_t \frac{1}{e_{ir} - b_{ir}} \sum_{t=b_{ir}}^{e_{ir}-1} \mathbf{h}_{it}\right) \in \mathbb{R}^d. \tag{14}$$

For phrase p_{ir} and image I_j, we first form the vector of phrase–patch similarities

$$\mathbf{z}_{ijr} = [z_{ijr1}, \ldots, z_{ijrN_p}]^\top \in \mathbb{R}^{N_p}, \qquad z_{ijrm} = \mathbf{u}_{ir}^\top \mathbf{v}_{jm}. \tag{15}$$

The phrase–image grounding score is the strongest patch response,

$$a_{ijr} = a(I_j, p_{ir}) = \max_{1 \le m \le N_p} z_{ijrm} \in \mathbb{R}. \tag{16}$$

Hence, $a(I_j, p_{ir})$ is a scalar associated with one phrase and one image; the maximization is only over the patches of that image. PPG evaluates every valid phrase of caption i against every image j in the current mini-batch. The local caption–image similarity is obtained by averaging the corresponding phrase-level grounding scores,

$$s_{ij}^{\ell} = s_{\ell}(I_j, T_i) = \frac{1}{P_i} \sum_{r=1}^{P_i} a_{ijr} \in \mathbb{R}. \tag{17}$$

Collecting these scalar entries gives the local similarity matrix

$$\mathbf{S}^{\ell} = [s_{ij}^{\ell}]_{i,j=1}^{B} \in \mathbb{R}^{B \times B}. \tag{18}$$

We apply the same identity-aware bidirectional distribution-matching principle to $\mathbf{S}^{\ell}$ and $\mathbf{Q}$, yielding the PPG objective $\mathcal{L}_{\mathrm{ppg}}$. The pre-max vector $\mathbf{z}_{ijr}$ also provides a direct visualization of phrase-specific evidence. If the image encoder outputs a spatial patch grid of size $H_p \times W_p$ with $N_p = H_p W_p$, the entries of $\mathbf{z}_{ijr}$ can be reshaped to that grid and upsampled to form a phrase grounding heatmap. This visualization depends only on the symbolic patch grid and is not tied to a particular input resolution.

3.5. Counterfactual Attribute Verification (CAV)

PPG exposes local visual evidence, but grounding alone does not require a model to reject a minimally corrupted description. CAV therefore converts each controlled counterfactual into two linked verification constraints. Unlike the batch-level PPG computation, CAV compares an original/corrupted text pair only with its own paired image I_i.

CAV-G: global verification. For selected training indices collected in $\mathcal{C} \subseteq \{1, \ldots, B\}$, define

$$s_i^{+} = s_{\mathrm{g}}(I_i, T_i), \qquad s_{ik}^{-} = s_{\mathrm{g}}(I_i, \widetilde{T}_{ik}). \tag{19}$$

Given a positive global margin $m_{\mathrm{g}} \in \mathbb{R}_{>0}$, CAV-G is

$$\mathcal{L}_{\mathrm{cav}}^{\mathrm{g}} = \frac{1}{|\mathcal{C}|K} \sum_{i \in \mathcal{C}} \sum_{k=1}^{K} \left[m_{\mathrm{g}} - s_i^{+} + s_{ik}^{-}\right]_{+}, \tag{20}$$

where $[x]_{+} = \max(0, x)$. This requires the correct caption to exceed each of its counterfactuals by at least m_{g}.

CAV-L: local verification. Encoding $\widetilde{T}_{ik}$ with the same text encoder yields a normalized embedding

$$\widetilde{\mathbf{u}}_{ik} \in \mathbb{R}^d \tag{21}$$

for its changed phrase $\widetilde{p}_{i,c_{ik}}$. The corresponding local supports on the paired image are

$$a_{ik}^{+} = a(I_i, p_{i,c_{ik}}), \qquad a_{ik}^{-} = a(I_i, \widetilde{p}_{i,c_{ik}}) \in \mathbb{R}. \tag{22}$$

Using a positive local margin $m_{\ell} \in \mathbb{R}_{>0}$, CAV-L is

$$\mathcal{L}_{\mathrm{cav}}^{\ell} = \frac{1}{|\mathcal{C}|K} \sum_{i \in \mathcal{C}} \sum_{k=1}^{K} \left[m_{\ell} - a_{ik}^{+} + a_{ik}^{-}\right]_{+}. \tag{23}$$

Thus, each counterfactual induces one local comparison between the original changed phrase and its corrupted counterpart, while each phrase support is itself computed over all patches of the same paired image. The complete verification objective is

$$\mathcal{L}_{\mathrm{cav}} = \mathcal{L}_{\mathrm{cav}}^{\mathrm{g}} + \mathcal{L}_{\mathrm{cav}}^{\ell}. \tag{24}$$

CAV-G tests whether the semantic corruption lowers global compatibility, whereas CAV-L ties that decrease to the exact changed attribute by requiring its corrupted phrase to lose local visual support. This coupling is the problem-specific mechanism that distinguishes GARNET from treating a counterfactual caption merely as a generic hard negative.

3.6. Auxiliary Objectives

We retain masked language modeling (MLM) as a complementary cross-modal relation-learning objective, following its effectiveness in TBPR [5]. A separate masked copy of T_i is passed to a cross-modal fusion module whose queries are text tokens and whose keys/values are the visual global and patch representations. Attribute-span tokens are masked with probability p_{attr}, while other eligible tokens are masked with probability p_{other}, with $p_{\mathrm{attr}} > p_{\mathrm{other}}$. Their numerical values are treated as implementation hyperparameters. MLM is not applied to a CAV-corrupted caption in the same forward path, preventing reconstruction from counteracting

the counterfactual verification signal. The resulting loss is $\mathcal{L}_{\text{mlm}}$. For identity supervision, let C be the number of training identities and

$$\mathbf{W}_I, \mathbf{W}_T \in \mathbb{R}^{C\times d} \tag{25}$$

denote the image and text identity classifiers. Using BNNeck-normalized global features, we define

$$\mathcal{L}_{\text{id}} = \text{CE}\big(\mathbf{W}_I\,\text{BN}(\mathbf{f}_i^I), y_i\big) + \text{CE}\big(\mathbf{W}_T\,\text{BN}(\mathbf{f}_i^T), y_i\big)\,. \tag{26}$$

The pre-BN normalized global features remain the retrieval representations.

3.7. Domain-Adaptive Pre-Training

Person-domain pre-training is used as a representation-strengthening stage rather than as the core contribution. We pre-train the same dual encoder and cross-modal MLM block on MALS [15]. With an MLM weight $\lambda_{\text{mlm}}^A \in \mathbb{R}_{\geq 0}$, the Stage-A objective is

$$\mathcal{L}_A = \mathcal{L}_{\text{sdm}} + \lambda_{\text{mlm}}^A \mathcal{L}_{\text{mlm}}. \tag{27}$$

PPG and CAV are introduced during benchmark-specific fine-tuning. Separating this stage from the proposed grounded verification mechanism allows gains from person-domain representation learning to be disentangled experimentally from the gains produced by PPG and CAV.

3.8. Fine-Tuning and Inference

During benchmark fine-tuning, all encoder parameters are optimized jointly. The full objective is

$$\mathcal{L}_B = \mathcal{L}_{\text{sdm}} + \lambda_{\text{ppg}}\mathcal{L}_{\text{ppg}} + \lambda_{\text{cav}}\mathcal{L}_{\text{cav}} + \lambda_{\text{mlm}}\mathcal{L}_{\text{mlm}} + \lambda_{\text{id}}\mathcal{L}_{\text{id}}, \tag{28}$$

where all loss weights are non-negative hyperparameters.

Implementation details. Complete preprocessing, architecture, optimization, and training settings are provided in the supplementary material.

4. Experiments and Results

4.1. Evaluation Protocol

We evaluate standard retrieval on CUHK-PEDES [7], ICFG-PEDES [3], and RSTPReid [19], reporting R@1/R@5/R@10 and mAP. Controlled diagnostics are evaluated on CUHK-PEDES unless stated otherwise.

For image I_i, caption T_i, its k-th controlled corruption $\widetilde{T}_{ik}$, and changed phrase pair $(p_{i,c_{ik}}, \widetilde{p}_{i,c_{ik}})$, let $\mathbb{E}_{i,k}[\cdot]$ denote the empirical average over samples and corruptions. We measure

$$\text{Acc}_{\text{g}}^{\text{C>C}} = \mathbb{E}_{i,k}\Big[\mathbf{1}\Big(s_{\text{g}}(I_i, T_i) > s_{\text{g}}(I_i, \widetilde{T}_{ik})\Big)\Big]\,, \tag{29}$$

$$\Delta_{\text{g}} = \mathbb{E}_{i,k}\Big[s_{\text{g}}(I_i, T_i) - s_{\text{g}}(I_i, \widetilde{T}_{ik})\Big]\,, \tag{30}$$

$$\text{Acc}_{\ell}^{\text{C>C}} = \mathbb{E}_{i,k}[\mathbf{1}(a(I_i, p_{i,c_{ik}}) > a(I_i, \widetilde{p}_{i,c_{ik}}))]\,, \tag{31}$$

$$\Delta_{\ell} = \mathbb{E}_{i,k}[a(I_i, p_{i,c_{ik}}) - a(I_i, \widetilde{p}_{i,c_{ik}})]\,. \tag{32}$$

Positive Δ_ℓ means that local evidence favors the correct phrase.

Attribute-confusable gallery. Attr-Conf. R@1 uses different-ID negatives that remain semantically close to the query but conflict in at least one visually verifiable attribute. Candidates are lexical-overlap filtered, ranked by a frozen method-independent Sentence-BERT encoder [11] (`all-mpnet-base-v2`), and the top-8 are retained after a method-independent annotation/visual sanity check. Full mining rules and statistics are provided in the supplement.

Prior-method local probe. Because IRRA, RaSa, MARS++, and CECA do not natively define our local metric, we evaluate their frozen checkpoints with a common evaluation-only probe that uses each method's existing shared embedding space for patch tokens and the changed phrase span, followed by the same max-over-patch similarity used by GARNET. No baseline is retrained or modified; GARNET uses its trained PPG score. Probe details are in the supplement.

Paraphrase robustness. We use the same deterministic rule-based, meaning-preserving paraphrase procedure as during training; generation details are provided in the supplement. Headline results use the primary global retrieval score without local re-ranking.

Standard retrieval. Table 1 compares GARNET with recent TBPR methods using the standard R@1, R@5, R@10, and mAP metrics on all three benchmarks.

GARNET achieves the best mAP on all three datasets and the best R@1 on ICFG-PEDES and RSTPReid among the values shown, while reaching 79.15 R@1 on CUHK-PEDES. It also attains the best reported R@10 on all three benchmarks and the best reported R@5 on CUHK-PEDES and RSTPReid. Thus, the verification objective preserves strong standard retrieval performance; we next test the targeted grounded behavior.

Controlled attribute verification. We compare representative global (IRRA [5]), perturbation-sensitive (RaSa [1]), attribute-aware (MARS++ [4]), and fine-grained (CECA [20]) baselines under the same controlled protocol.

Table 2 exposes the targeted global–local gap. Prior methods remain strong under Attr-Conf. retrieval and usually prefer the correct caption globally, yet under the common evaluation-only local probe the changed phrase is separated much less reliably; RaSa and MARS++ even yield negative Δ_ℓ. CECA is the strongest evaluated prior local baseline at

Table 1. **Standard TBPR comparison on three benchmarks.** R@1, R@5, R@10, and mAP are reported in percentages; **bold**/underline denote best/second-best values. † denotes MALS person-domain pre-training.

Method	Venue	CUHK-PEDES				ICFG-PEDES				RSTPReid			
		R@1	R@5	R@10	mAP	R@1	R@5	R@10	mAP	R@1	R@5	R@10	mAP
CLIP-ViT-B/16 [10]	–	68.19	86.47	91.47	61.12	56.74	75.72	82.26	31.84	54.05	80.70	88.00	43.41
IRRA [5]	CVPR'23	73.38	89.93	93.71	66.13	63.46	80.25	85.82	38.06	60.20	81.30	88.20	47.17
RaSa [1]	IJCAI'23	76.51	90.29	94.25	69.38	65.28	80.40	85.12	41.29	66.90	86.50	91.35	52.31
APTM† [15]	ACM MM'23	76.53	90.04	94.15	66.91	68.51	82.99	87.56	41.22	67.50	85.70	91.45	52.56
RDE [8]	CVPR'24	75.94	90.14	94.12	67.56	67.68	82.47	87.36	40.06	65.35	83.95	89.90	50.88
CFAM [21]	CVPR'24	75.60	90.53	94.36	67.27	65.38	81.17	86.35	39.42	62.45	83.55	91.10	49.50
MARS++† [4]	TOMM'25	77.89	90.68	93.91	71.46	68.50	82.29	86.43	45.69	67.80	86.65	91.55	53.23
IAPTM (HCA) [2]	ACM MM'25	77.44	91.23	94.61	68.92	66.95	82.34	87.06	40.13	69.70	86.95	92.45	53.72
ICL [9]	CVPR'25	77.91	90.27	94.14	69.13	69.02	82.45	87.36	41.21	70.55	85.95	91.65	53.68
KPDM-L/14 [17]	AAAI'26	78.02	91.49	95.00	69.71	69.13	83.46	88.10	42.78	67.95	86.35	91.65	52.97
ALIGNER [16]	TIFS'26	76.20	90.81	94.61	68.00	68.61	82.81	87.44	41.07	68.65	85.00	90.15	51.74
CADA+QC-Align [6]	CVPR'26	**79.31**	91.86	95.17	–	68.79	83.16	87.81	–	71.10	87.50	92.75	–
CECA [20]	CVPR'26	78.30	91.10	94.09	65.92	72.25	**85.31**	89.32	37.57	71.40	88.30	93.35	52.07
GARNET† (ours)	–	79.15	**92.04**	**95.45**	**71.95**	**72.34**	83.63	**89.95**	**48.26**	**73.06**	**89.30**	**93.55**	**55.81**

Table 2. **Controlled attribute verification on CUHK-PEDES.** Global C>C evaluates caption-level rejection; Local C>C and Δ_ℓ evaluate the exact changed phrase. ‡ denotes the evaluation-only post-hoc local probe on frozen prior checkpoints; GARNET uses its trained PPG score.

Method	Representative capability	Attr-Conf. R@1 ↑	Global C>C (%) ↑	Δ_g ↑	Local C>C (%) ↑	Δ_ℓ ↑
IRRA [5]	Global + implicit relation alignment	93.26	86.73	0.055	$58.14^{‡}$	$0.008^{‡}$
RaSa [1]	Replaced-word / perturbation sensitivity	94.38	86.21	0.058	$41.40^{‡}$	$-0.012^{‡}$
MARS++ [4]	Attribute-aware learning	95.45	84.28	0.065	$34.63^{‡}$	$-0.015^{‡}$
CECA [20]	Fine-grained attribute correspondence	93.26	85.95	0.051	$81.52^{‡}$	$0.034^{‡}$
GARNET	**Grounded counterfactual verification**	**95.51**	**92.04**	0.034	**91.92**	**0.135**

81.52% Local C>C and $\Delta_\ell = 0.034$, whereas GARNET reaches 91.92% and 0.135 (+10.40 points in Local C>C). This supports the central claim: strong global alignment, perturbation sensitivity, or fine-grained correspondence does not by itself guarantee grounded verification of the exact changed phrase.

Semantic preservation. GARNET changes from 79.15 R@1 on original queries to 79.18 under meaning-preserving paraphrases ($\Delta_{\text{para}} = -0.03$), consistent with robustness to lexical rewording. Full results are in the supplement.

Qualitative analysis. We retain one task-level attribute-confusable retrieval example in the main paper; additional qualitative analyses are provided in the supplement.

Figure 3 illustrates the task-level effect of the verification gap. Competing methods retrieve different identities that preserve several query attributes but remain inconsistent in a decisive detail, whereas GARNET ranks the ground-truth identity first. This qualitatively complements the Attr-Conf. R@1 diagnostic in Table 2.

Overall, GARNET remains competitive on standard TBPR benchmarks, while the controlled comparison exposes the prior-work verification gap and Sec. 5 shows that PPG/CAV directly target it. Additional robustness and specificity analyses are provided in the supplement.

Table 3. **Core GARNET ablation.** "N/A" denotes variants without a native phrase-grounding score.

Configuration	R@1 ↑	mAP ↑	Attr-Conf. R@1 ↑	Global C>C ↑	Local C>C ↑	Δ_ℓ ↑
Baseline	74.40	65.04	88.26	85.82	N/A	N/A
Stage A	76.37	66.96	90.38	86.43	N/A	N/A
Stage A + PPG	78.15	68.64	91.38	89.95	88.05	0.106
Stage A + CAV-G	78.45	69.04	92.05	90.72	N/A	N/A
Stage A + PPG + CAV-G	77.84	68.65	94.38	90.46	89.86	0.115
Full GARNET (+CAV-L)	**79.15**	**71.95**	**95.51**	**92.04**	**91.92**	**0.135**

5. Ablation and Mechanism Analysis

Table 2 identifies a global–local verification gap in the evaluated prior methods. We ablate GARNET to test how PPG/CAV target this gap beyond stronger initialization or generic perturbation training.

Core component ablation. We compare Stage A and complementary PPG/CAV configurations to isolate their contributions to retrieval and grounded verification.

Stage A mainly improves retrieval, PPG introduces local phrase evidence, and CAV-G strengthens global rejection. Adding CAV-L to PPG+CAV-G raises Local C>C from 89.86% to 91.92% and Δ_ℓ from 0.115 to 0.135, while producing the strongest overall results. The gain is therefore concentrated on the grounded local behavior targeted

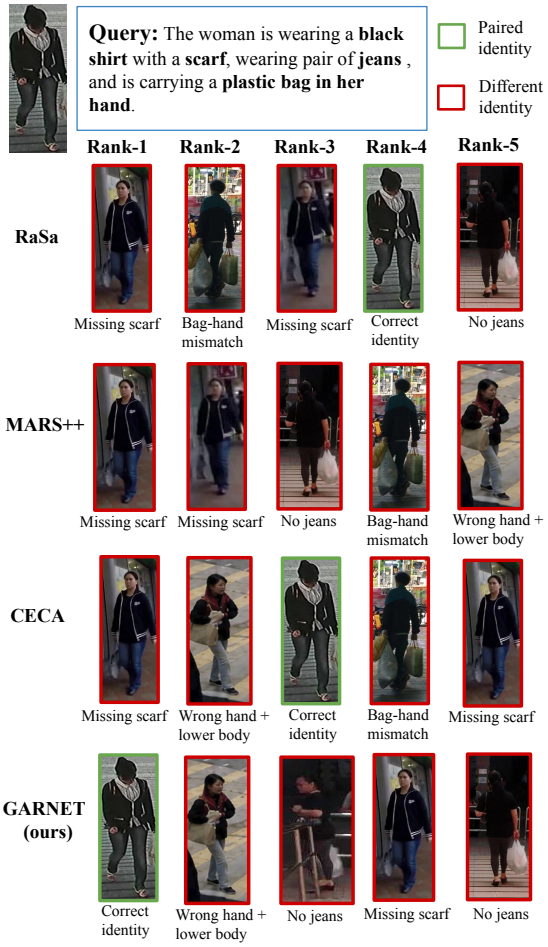


Figure 3. **Attribute-confusable retrieval.** Top-5 results for RaSa [1], MARS++ [4], CECA [20], and GARNET on the same query. Green borders denote the correct identity and red borders denote incorrect identities; short tags indicate the decisive query-relevant mismatch. Prior methods rank attribute-confusable distractors above the correct identity, whereas GARNET retrieves the correct identity at Rank-1.

Table 4. **Generic perturbation vs. controlled semantic corruption.** Local verification is reported in Table 3.

Supervision / variant	Attr-Conf. R@1 ↑	Global C>C ↑	Δ_g ↑
Random caption negative	90.49	83.18	0.016
Random token replacement	91.38	84.95	0.015
Single-attribute corruption + CAV-G	92.05	90.72	0.029
PPG + CAV-G	94.38	90.46	0.026
Full GARNET (+CAV-L)	**95.51**	**92.04**	**0.034**

by GARNET.

Generic perturbation vs. controlled verification. We next test whether generic negatives can provide the same benefit as controlled single-attribute corruption under the same training setup.

Generic caption/token perturbations yield weaker Global C>C than the controlled single-attribute corruption. Adding PPG and the full CAV objective further improves Attr-Conf. R@1 to 95.51% and Global C>C to 92.04%, consistent with the benefit of targeted semantic counterfactual supervision over generic hard negatives alone. Changed-phrase specificity and corruption-type analyses are provided in the supplementary material.

Together, the ablations provide a component-wise progression from the identified prior-work gap to GARNET: PPG exposes phrase evidence, CAV-G verifies the contradiction globally, and CAV-L verifies the exact changed phrase locally.

6. Conclusion

We presented GARNET for text-based person retrieval, motivated by a specific gap between existing fine-grained correspondence and perturbation-sensitive learning. Prior TBPR methods can model local attributes or detect textual changes, yet these capabilities do not necessarily require the local evidence of the *exact changed phrase* to decrease when that phrase becomes visually inconsistent. GARNET addresses this gap by coupling Phrase–Patch Grounding with Counterfactual Attribute Verification: PPG exposes phrase-specific evidence, CAV-G verifies controlled contradictions globally, and CAV-L explicitly verifies the changed phrase locally.

Across CUHK-PEDES, ICFG-PEDES, and RSTPReid, GARNET remains competitive under standard retrieval evaluation. The controlled CUHK-PEDES diagnostics further show the behavior targeted by the method: GARNET reaches 91.92% Local C>C with $\Delta_\ell = 0.135$, compared with 81.52% and 0.034 for the strongest evaluated prior local baseline. The ablations show that these gains are not explained by person-domain pre-training or generic lexical/hard negatives alone, but are concentrated on the grounded correct-over-corrupt verification behavior introduced by PPG and CAV. These results support grounded attribute verification as a useful complement to conventional global and fine-grained alignment in TBPR.

Societal Impact

Text-based person retrieval can support applications such as missing-person search and large-scale image organization, but it can also be used in privacy-invasive or surveillance-oriented settings. GARNET improves how a retrieval model verifies visually grounded attributes; it does not by itself resolve questions of consent, demographic bias, data governance, or appropriate deployment. Practical use should therefore follow applicable privacy and data protection requirements, evaluate performance across relevant populations, and include human oversight when retrieval outcomes may affect individuals.

References

[1] Yang Bai, Min Cao, Daming Gao, Ziqiang Cao, Chen Chen, Zhenfeng Fan, Liqiang Nie, and Min Zhang. RaSa: Relation and sensitivity aware representation learning for text-based person search. In *Proceedings of the Thirty-Second International Joint Conference on Artificial Intelligence*, pages 555–563. International Joint Conferences on Artificial Intelligence Organization, 2023.

[2] Yifei Deng, Chenglong Li, Futian Wang, and Jin Tang. Learning hierarchical cross-modal association with intra-modal context for text-image person retrieval. In *Proceedings of the 33rd ACM International Conference on Multimedia*, pages 2723–2731, 2025.

[3] Zefeng Ding, Changxing Ding, Zhiyin Shao, and Dacheng Tao. Semantically self-aligned network for text-to-image part-aware person re-identification. *arXiv preprint arXiv:2107.12666*, 2021.

[4] Alex Ergasti, Tomaso Fontanini, Claudio Ferrari, Massimo Bertozzi, and Andrea Prati. MARS: Paying more attention to visual attributes for text-based person search. *ACM Transactions on Multimedia Computing, Communications, and Applications*, 21(10), 2025.

[5] Ding Jiang and Mang Ye. Cross-modal implicit relation reasoning and aligning for text-to-image person retrieval. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 2787–2797, 2023.

[6] Dongsheng Li, Xinyuan Guo, Huijie Zhang, Pingting Hao, and Qiushi Xia. Quota-calibrated fine-grained alignment with context-aware marginals for text-based person retrieval. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 31103–31112, 2026.

[7] Shuang Li, Tong Xiao, Hongsheng Li, Bolei Zhou, Dayu Yue, and Xiaogang Wang. Person search with natural language description. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 1970–1979, 2017.

[8] Yang Qin, Yingke Chen, Dezhong Peng, Xi Peng, Joey Tianyi Zhou, and Peng Hu. Noisy-correspondence learning for text-to-image person re-identification. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 27197–27206, 2024.

[9] Yang Qin, Chao Chen, Zhihang Fu, Dezhong Peng, Xi Peng, and Peng Hu. Human-centered interactive learning via MLLMs for text-to-image person re-identification. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 14390–14399, 2025.

[10] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. Learning transferable visual models from natural language supervision. In *Proceedings of the 38th International Conference on Machine Learning*, pages 8748–8763. PMLR, 2021.

[11] Nils Reimers and Iryna Gurevych. Sentence-BERT: Sentence embeddings using siamese BERT-networks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing*, pages 3982–3992. Association for Computational Linguistics, 2019.

[12] Zhiyin Shao, Xinyu Zhang, Meng Fang, Zhifeng Lin, Jian Wang, and Changxing Ding. Learning granularity-unified representations for text-to-image person re-identification. In *Proceedings of the 30th ACM International Conference on Multimedia*, pages 5566–5574, 2022.

[13] Xiujun Shu, Wei Wen, Haoqian Wu, Keyu Chen, Yiran Song, Ruizhi Qiao, Bo Ren, and Xiao Wang. See finer, see more: Implicit modality alignment for text-based person retrieval. In *European Conference on Computer Vision Workshops*, pages 624–641, 2022.

[14] Zhe Wang, Zhiyuan Fang, Jun Wang, and Yezhou Yang. ViTAA: Visual-textual attributes alignment in person search by natural language. In *Proceedings of the European Conference on Computer Vision*, pages 402–420, 2020.

[15] Shuyu Yang, Yinan Zhou, Yaxiong Wang, Yujiao Wu, Li Zhu, and Zhedong Zheng. Towards unified text-based person retrieval: A large-scale multi-attribute and language search benchmark. In *Proceedings of the 31st ACM International Conference on Multimedia*, pages 4492–4501, 2023.

[16] Xi Yang, Kun Chen, and Nannan Wang. ALIGNER: Learning fine-grained cross-modal alignment for text-based person retrieval. *IEEE Transactions on Information Forensics and Security*, 21:4726–4740, 2026.

[17] Shaofeng You, Tianle Miao, Qihang Chen, Xin Li, Zhuo Cheng, and Dapeng Luo. KPDM: Key phrase dynamic masking for robust text-to-image person retrieval. In *Proceedings of the AAAI Conference on Artificial Intelligence*, pages 12099–12107, 2026.

[18] Ying Zhang and Huchuan Lu. Deep cross-modal projection learning for image-text matching. In *Proceedings of the European Conference on Computer Vision*, pages 686–701, 2018.

[19] Aichun Zhu, Zijie Wang, Yifeng Li, Xili Wan, Jing Jin, Tian Wang, Fangqiang Hu, and Gang Hua. DSSL: Deep surroundings-person separation learning for text-based person retrieval. In *Proceedings of the 29th ACM International Conference on Multimedia*, pages 209–217, 2021.

[20] Hao Zou, Runqing Zhang, Jin Ding, Xue Zhou, Jianxiao Zou, and Mingzhu Cai. Tackling alignment ambiguity in person retrieval through conversational attribute mining. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 9689–9698, 2026.

[21] Jialong Zuo, Hanyu Zhou, Ying Nie, Feng Zhang, Tianyu Guo, Nong Sang, Yunhe Wang, and Changxin Gao. UFineBench: Towards text-based person retrieval with ultra-fine granularity. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 22010–22019, 2024.